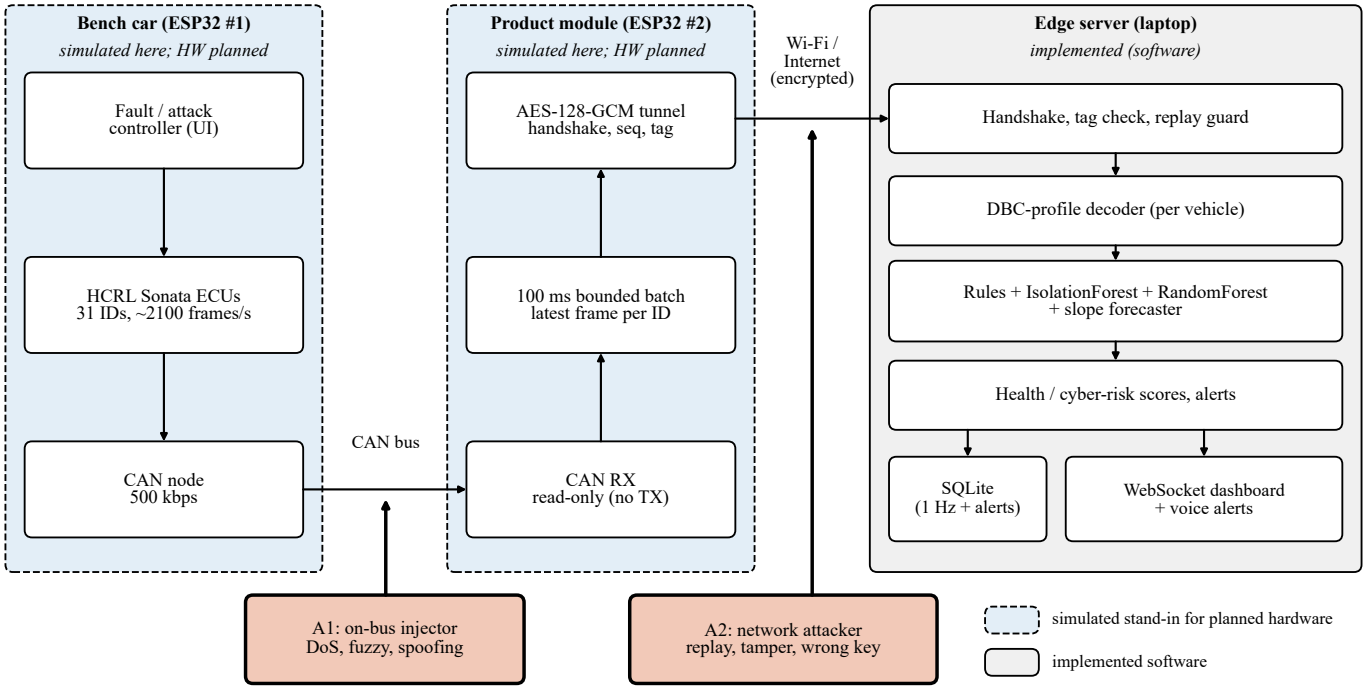

Bench car (ESP32 #1)
simulated here; HW planned
Fault / attack controller (UI)
HCRL Sonata ECUs 31 IDs, ~2100 frames/s
CAN node 500 kbps
CAN bus
Product module (ESP32 #2)
simulated here; HW planned
CAN RX read-only (no TX)
100 ms bounded batch latest frame per ID
AES-128-GCM tunnel handshake, seq, tag
Wi-Fi / Internet (encrypted)
Edge server (laptop)
implemented (software)
Handshake, tag check, replay guard
DBC-profile decoder (per vehicle)
Rules + IsolationForest + RandomForest + slope forecaster
Health / cyber-risk scores, alerts
SQLite (1 Hz + alerts)
WebSocket dashboard + voice alerts
A1: on-bus injector DoS, fuzzy, spoofing
A2: network attacker replay, tamper, wrong key
simulated stand-in for planned hardware
implemented software